Batch 7 - Analysis sheet: exam timetabling (graph colouring)

Chosen: **Backtracking** + forward checking + most-constrained-first. Instance: chi = 3; timetable slot1 {DAA, CN, Maths}, slot2 {OS, ML, SE}, slot3 {DBMS}.

A. Problem types

Question	Type
Fit in 3 slots?	Decision (NP-complete)
Minimum slots?	Optimisation (NP-hard)
List all timetables	Enumeration (36 labelled / 6 distinct)

B. Data structures

Operation	Adjacency sets (chosen)	Adjacency matrix	Edge list
Space	O(n+m)	O(n^2)	O(m)
Conflict test	O(1) avg	O(1)	O(m)
Neighbours of v	O(deg v)	O(n)	O(m)
Domains	bitmask: remove/empty test O(1)	-	-

C. Techniques compared

Technique	Verdict	Concrete evidence
Backtracking	CHOSEN	Local prunable constraint; exact; 0 backtracks on real-size data (n=420, 30 ms); worst case O(k^n n)
Greedy	Rejected	8-vertex tree: descending-degree/lowest-slot uses 3, optimum 2
Dynamic programming	Rejected	Subset states 2^n, O(3^n): 197 ms at n=14, 1,837 ms at n=16, x9 per +2; n=420 impossible
Divide and conquer	Rejected	Cross edges couple halves; merge needs recolouring; no cheap merge
Branch and bound	Rejected	Fixed k: no objective; clique bound = 2 on Mycielski graphs with chi up to 6 -> no pruning

D. Complexity

	Time	Space
Backtracking worst	O(k^n * n)	O(n^2) as coded (O(n+m) with undo trail)
Backtracking best	O(n^2 + m)	
Subset DP	O(3^n)	O(2^n)

E. Measured (random graphs; see results/*.csv)

n	6	8	10	12	14	16
Backtracking ms	0.03	0.05	0.06	0.07	0.09	0.11
Subset DP ms	0.05	0.23	2.2	21.5	197	1,837

Worst case (Mycielski, k-1 slots): nodes 26 / 895 / 448,614 for chi = 4 / 5 / 6 (most-constrained-first); input order 57 / 49,388 / >3,000,000.

F. Changed requirement (Maths slot 1, one exam per day)

Colour by **day** (9 colours; second slot only balances load). Maths: allowed = {D1}. Ordering: most-constrained-first (pinned course first, then smallest domain, ties by conflicts) - n nodes vs up to millions for input order. 3 slots (= 2 days) infeasible: the DAA-OS-DBMS triangle needs 3 days; 3 days: Maths 1, DAA 1, CN 2, OS 3, SE 3, ML 4, DBMS 5. Backtracking still applies (precolouring is NP-complete).